

Aaryan Patel

a52patel@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | aaryan-patel.com

EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

June 2027

TECHNICAL SKILLS

Languages: C/C++, JavaScript/TypeScript, SQL (PostgreSQL), Java, Scala, Python, Bash, Kotlin

Databases/Frameworks: .NET, FastAPI, Node.js, Mongo, Redis, Next.js, Fastify, Redux, Hadoop, Spark

DevOps: Git, Docker, AWS, Kubernetes, Prometheus, Terraform, Jenkins, GitHub Actions, Cloudflare

AI/Machine Learning: Pandas, PyTorch, Prompt Engineering, LangChain, Hugging Face, OpenAI API

EXPERIENCE

Software Engineer Intern

ZipHQ

Sep. 2026 – Dec. 2026

Toronto, ON

DevOps Engineer Intern

RBC Capital Markets

Sep. 2025 – Dec. 2025

Toronto, ON

- Engineered a **multi-cloud deployment architecture** for distributed app across Azure and AWS clusters by using **GitHub Actions** pipelines and **Helm charts** to orchestrate the build and release of Dockerized microservices
- Accelerated container security scanning speeds by over 77%** for Capital Markets teams by migrating legacy scanning protocols to Aqua SaaS Cloud, significantly reducing **Jenkins CI/CD pipeline latency**
- Enabled verification of Docker image signatures** for an application by provisioning Virtual Machines with custom **CA bundles**, ensuring only trusted code runs in production

Backend Developer

Scrawlr

May 2025 – Aug 2025

Vancouver, BC

- Contributed to scalable backend APIs** for browser-extension platform, using Node.js + Fastify with Postgres and Redis to layer real-time conversation and personalized content onto any webpage
- Standardized codebase by **authoring reusable Cursor rules** that enforce layered architecture, uniform error handling, and code consistency, enabling faster, conflict-free feature rollouts across teams
- Architected an AI-driven code-gen pipeline** that ingests product docs & Cursor rules to auto-scaffold Fastify services, database schemas, cache layers, and Vitest tests, **cutting API development from months to days**

Software Engineer Intern

RBC Capital Markets

Sep. 2024 – Dec. 2024

Toronto, ON

- Developed a multi-asset trading analytics platform using **Next.js**, **React**, **q & kdb+**, and **MongoDB**, on **Kubernetes**, integrating **AI-driven trade reports** enhance market prediction capabilities for traders
- Optimized client-side data handling** for datasets exceeding 20,000 rows by implementing **Redux** and **TanStack Query**, reducing API call latency by **40%** through aggressive caching and state management strategies
- Engineered high-performance visualization modules** (including FX Liquidity Matrix) using **Highcharts**, **AG Grid**, and **React Virtualized**, improving rendering speeds by **35%** via DOM virtualization

Software Developer

JANA Corporation

May 2023 – Aug. 2023

Aurora, ON

- Developed backend services** for a pipeline integrity management application using **C#**, **ASP.NET**, and **SQL Server**, implementing the data processing logic for predictive risk analysis and visualization features
- Boosted API performance by up to 50%** by refactoring endpoints to utilize native high-performance serialization libraries, significantly reducing latency and payload overhead
- Optimized data retrieval efficiency** for RESTful services by restructuring complex **SQL queries** and stored procedures, ensuring faster dynamic data fetching for downstream application

PROJECTS & LEADERSHIP

[Atrangi Eventz](#) | *Next.js, PostgreSQL, Redis, Stripe, Cloudflare, Amazon S3, SES*

- Engineered a cost-efficient ticketing platform in **Next.js & PostgreSQL**, integrating **Stripe** for secure payments
- Implemented **Redis-based distributed locking** to handle concurrent bookings, ensuring strict data integrity
- Managed media and emails with **AWS S3 & SES**, and built a **QR code system** for rapid attendee check-in

[UW Data Science Club](#) | *TypeScript, Next.js, PostgreSQL, Node.js, GitHub Actions*

- Led a **10-person team** and designed a **PostgreSQL** database supporting **750+ club members** and 60+ events
- Architected **agentic workflows** with quality gates, centralized secrets, & **automation suite** for repo onboarding
- Built a private **Node SDK** for **error tracking** while driving **monorepo adoption** to isolate admin operations